A proof of the conjectured linear recurrence and generating function for OEIS A253019

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A253019 as

> *Number of (n+2) X (2+2) 0..3 arrays with every consecutive three elements in every row and column not having exactly two distinct values, and in every diagonal and antidiagonal having exactly two distinct values, and new values 0 upwards introduced in row major order*

and carries in its Formula section the lines:

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Empirical:  a(n) = 6*a(n-3) -10*a(n-6) +3*a(n-9) for n>11.

Empirical g.f.:  x*(37 + 50*x + 83*x^2 - 131*x^3 - 167*x^4 - 275*x^5 +
67*x^6 + 59*x^7 + 87*x^8 - 6*x^9 - 3*x^10) / ((1 - 3*x^3)*(1 - 3*x^3 +
x^6)).  - Colin Barker, Dec 08 2018
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That line carries no separate signature; the entry itself is by R. H. Hardin (Dec 26 2014).

The line quoted ending in that signature is contributed by Colin Barker on Dec 08 2018.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 8, last modified Dec 08 2018, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 6\,a(n-3) - 10\,a(n-6) + 3\,a(n-9).$$

stated for $n > 11$.

Written out, the claim is

$$\sum_{n\geq 1} a(n)\,x^n \;=\; \frac{37x + 50x^2 + 83x^3 - 131x^4 - 167x^5 - 275x^6 + 67x^7 + 59x^8 + 87x^9 - 6x^{10} - 3x^{11}}{1 - 6x^3 + 10x^6 - 3x^9}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 4$ array over $\{0, \dots, 3\}$ with $n' = n + 2$ rows.

Every three such cells, with $(\delta, \varepsilon) \in \{(0, 1), (1, 0)\}$, must not carry exactly 2 distinct values.

Every three such cells, with $(\delta, \varepsilon) \in \{(1, -1), (1, 1)\}$, must carry exactly 2 distinct values.

The entry's labelling clause says that reading the array row by row the values $0, \dots, 3$ first occur in increasing order.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Every condition looks at three cells in a straight line, so it spans at most three consecutive rows. Taking as state the last two rows together with a counter for how many values have been introduced makes every condition decidable at the moment its lowest row is read.

An array of n' rows is then exactly a walk of length n' from the empty state, so $a(n) = w^{\mathsf{T}} M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 380 states, 380 after trimming; the forward identification of states with equal numbers of completions of every length, then the mirror identification on the edges coming in, leaves $S = 14$ blocks, and the count satisfies the linear recurrence given by the characteristic polynomial of that 14×14 matrix. The bound is computed, not assumed.

5 The decision procedure

Let $c_1, \dots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{\,m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 14: writing that polynomial as $t^{14} - \sum_{i=1}^{14} \lambda_i t^{14-i}$ gives $u_{m+14} = \sum_{i=1}^{14} \lambda_i u_{m+14-i}$. Hence 14 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A253019, the recurrence $a(n) = 6\,a(n-3) - 10\,a(n-6) + 3\,a(n-9)$ holds for every $n > 11$, and fails at $n = 11$.*

Proof. The residuals u_m were computed exactly for $m = 1, \ldots, 66$. The last nonvanishing one is at $m = 2$, that is at $n = 11$. After it, more than $S = 14$ consecutive residuals vanish, so by Section 5 every later residual vanishes as well. □

The entry claims the recurrence for $n > 11$. The theorem is at least as strong.

Theorem 2. *For A253019, the generating function is exactly $\sum_{n\geq 1} a(n)x^n = \frac{37x+50x^2+83x^3-131x^4-167x^5-275x^6+67x^7}{1-6x^3+10x^6-3x^9}$*

Proof. Let $N(x)$ and $D(x)$ be the numerator and denominator above, $D(0) = 1$. The residual test applied to the recurrence read off D — namely $a(n) = \sum_i c_i a(n-i)$ with c_i the negated coefficients of D — gives vanishing residuals for every $n > 11$, so the power series $F(x) = \sum_{n\geq 1} a(n)x^n$ satisfies $[x^k]\big(F(x)D(x)\big) = 0$ for every $k > 12$. The remaining coefficients of $F(x)D(x)$, for $k \leq 23$, were computed from the walk count and agree with $N(x)$ term by term. Hence $F(x)D(x) = N(x)$ identically, which is the assertion. □

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 37 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 37, 50, 83, 91, 133, 223, 243, 357, …
- **An independent enumeration.** For $n = 1, \ldots, 2$ every one of the 4^{4n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 14$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A253019, revision 8, last modified Dec 08 2018.

2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.

3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.

4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.